

Project: AI Learning Coach – RAG Chatbot

Overview: The AI Learning Coach is a Retrieval-Augmented Generation (RAG) application that allows users to upload PDF documents and chat with them using natural language. The chatbot retrieves relevant information from uploaded documents before generating responses, ensuring accurate and context-aware answers.

Project Objective

- Upload multiple PDF documents
- Read and process PDF content
- Generate embeddings for document chunks
- Store embeddings in a vector database
- Retrieve relevant information using semantic search
- Answer user questions based only on uploaded documents

Learning Outcomes

- Understand Retrieval-Augmented Generation (RAG)
- Document loading and preprocessing
- Text chunking
- Embeddings and vector databases
- Semantic search
- LangChain tools
- LangGraph agents
- Streamlit application development

Technology Stack

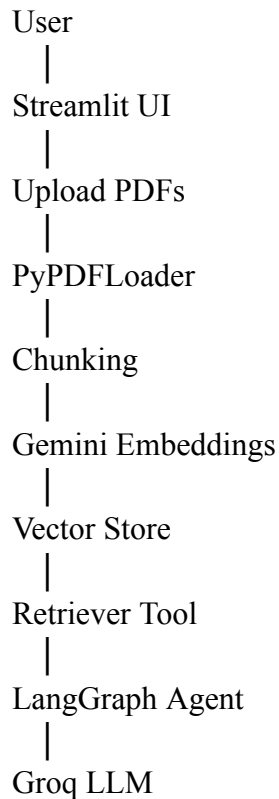
Python
Streamlit
LangChain

LangGraph
Google Gemini Embeddings
Groq LLM (GPT-OSS-20B)
PyPDFLoader
RecursiveCharacterTextSplitter
InMemoryVectorStore
python-dotenv

System Workflow

1. User uploads PDF files.
2. PyPDFLoader extracts document text.
3. RecursiveCharacterTextSplitter divides text into chunks.
4. Gemini Embeddings converts chunks into vectors.
5. InMemoryVectorStore stores vectors.
6. Retriever Tool performs similarity search.
7. LangGraph Agent retrieves context.
8. Groq LLM generates answers using retrieved context.
9. Streamlit displays responses to the user.

Architecture



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Final Response

Features

- Multiple PDF Upload
- Semantic Search
- AI Question Answering
- Conversation Memory
- Streamlit Chat Interface
- Retrieval-Based Responses
- Easy Integration with Additional Documents

Folder Structure

```
AI_RAG_CHATBOT/  
├── app.py  
├── .env  
├── requirements.txt  
├── doc_files/  
├── assets/  
├── utils/  
└── README.md
```

Future Enhancements

- Replace InMemoryVectorStore with ChromaDB or FAISS
- Add source citations
- Support DOCX, PPTX and Excel
- Export chat history
- Voice interaction
- User authentication

Conclusion

This project demonstrates the complete RAG pipeline, including document ingestion, chunking, embeddings, vector search, retrieval, and answer generation using an AI agent. It serves as an excellent capstone project for understanding modern Generative AI applications.

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